

Vietnamese Legal Question Answering via Cross-Encoder Reranking and Fine-Tuned Reasoning: Architecture, Empirical Evaluation, and Audit of Citation Quality

Anonymous

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Abstract

Legal Question Answering (LegalQA) poses distinct technical challenges compared to general text QA, as answers must be strictly grounded in valid statutory authority, respect temporal applicability, and provide verifiable legal citations. In this work, we present an architectural framework and comprehensive empirical evaluation of a Vietnamese LegalQA system based on Retrieval-Augmented Generation (RAG) and Large Language Model (LLM) reasoning. Our pipeline indexes a corpus of 643,469 legal text chunks spanning 261,440 unique citation identifiers, incorporating a multi-stage retrieval system enhanced by a cross-encoder reranker trained on 357,044 negative-sampled pairs and a QLoRA fine-tuned Qwen2.5-7B-Instruct generator.

Our empirical evaluation on 300 test queries demonstrates that model-based cross-encoder reranking yields substantial retrieval gains, elevating Recall@5 to 64.67% and Mean Reciprocal Rank (MRR) to 0.4828 compared to BM25 baseline (MRR 0.3370). For generation, QLoRA instruction tuning achieves superior answer overlap (Token F1 0.5909, ROUGE-L 0.4808) in oracle context. Crucially, a data leakage audit confirms zero exact test overlap with our 89,261 QA-memory records. Furthermore, a multi-axis human evaluation reveals a striking divergence: while automated proxies report near-perfect citation presence (1.00), human legal evaluators rate citation correctness at only 3.64/5 with a 36% failure rate, proving that automated metrics severely overestimate citation reliability. We outline key open challenges, including minimal-span retrieval, temporal validity filtering, and claim-level verification for trustworthy LegalQA.

1 Introduction

The deployment of Large Language Models (LLMs) and Retrieval-Augmented Generation (RAG) frameworks (Lewis et al., 2020) has transformed information access across domain-specific applications. In the legal domain, automated Question Answering (LegalQA) holds immense promise for streamlining legal research, assisting legal practitioners, and expanding Access to Justice (A2J) for citizens (Li et al., 2024; Abdallah et al., 2023). However, LegalQA imposes fundamentally different technical and normative constraints compared to general open-domain TextQA benchmarks such as SQuAD (Rajpurkar et al., 2018), Natural Questions (Kwiatkowski et al., 2019), or HotpotQA (Yang et al., 2018).

In general TextQA, an answer is deemed successful if it achieves high semantic or lexical overlap with ground-truth text snippets. In contrast, a legally valid answer must not only be textually accurate but must also establish statutory authority, cite specific legal provisions (e.g., Articles, Clauses, and Points), respect temporal applicability (distinguishing currently enforced law from repealed statutes), and account for jurisdictional hierarchy (Guha et al., 2023; Pipitone and Alami, 2024).

The consequences of ungrounded responses in LegalQA are severe. Recent empirical investigations demonstrate that commercial LLMs and RAG-based legal tools suffer from high

rates of "legal hallucination," frequently inventing fictitious judicial precedents or misapplying statutory clauses (Dahl et al., 2024; Magesh et al., 2024). Courts globally have reported a surge in fraudulent court filings containing synthetic case citations generated by uncalibrated AI tools. In response, professional bodies and regulatory frameworks—such as ABA Formal Opinion 512, NIST AI Risk Management Framework 600-1, and the EU AI Act—increasingly mandate verifiable proof of legal grounding and strict auditability for legal AI applications.

Building reliable LegalQA systems for the Vietnamese legal landscape presents unique domain challenges. Vietnamese legal documents follow a complex statutory hierarchy consisting of Codes, Laws, Decrees, Circulars, and Decisions, characterized by intricate cross-references and frequent periodic updates. While foundational benchmarks such as VLQA (Nguyen et al., 2025), DRILL (VLSP Consortium, 2025), and LawViVQA (Nguyen et al., 2024) have recently advanced legal information retrieval and visual document understanding in Vietnam, comprehensive end-to-end architectures that integrate multi-stage retrieval, model-based reranking, instruction-tuned generation, data leakage auditing, and human verification remain underdeveloped.

To address these challenges, this paper presents an end-to-end Vietnamese LegalQA framework and a rigorous empirical investigation into its retrieval, generation, leakage, and citation verification performance. Our major contributions are summarized as follows:

1. **Scalable End-to-End Pipeline:** We design a Vietnamese LegalQA pipeline indexing a massive legal corpus of 643,469 text chunks and 261,440 unique citation IDs. The framework integrates a hybrid retrieval module, a 2-stage cross-encoder reranker trained on 357,044 negative-sampled pairs, a QA-memory cache of 89,261 records, and a QLoRA fine-tuned Qwen2.5-7B-Instruct generator trained on 26,230 SFT examples.
2. **Empirical Retrieval and Generation Benchmarking:** We conduct extensive ablation studies evaluating retrieval variants across 300 test queries. We show that model-based cross-encoder reranking dramatically improves retrieval precision, boosting Recall@5 from 42.67% (BM25) to 64.67% and MRR from 0.3370 to 0.4828. For generation, QLoRA instruction tuning yields significant gains in answer fidelity (Token F1 0.5909, ROUGE-L 0.4808) over prompt-only baselines.
3. **Rigorous Data Leakage Audit:** We execute a character n -gram similarity audit comparing 1,458 test questions against 89,261 QA-memory records, confirming zero exact or normalized duplicates, thus guaranteeing that QA-memory retrieval gains do not stem from test data contamination.
4. **Human Evaluation and Proxy Metric Discrepancy:** We conduct a multi-axis human evaluation across 75 test instances (150 double-blind annotations). We uncover a critical empirical insight: while automatic citation presence proxies score 1.00, human legal evaluators rate citation correctness at only 3.64/5, with 36% of citations failing to support the specific legal claim. This highlights the urgent necessity of moving beyond lexical metrics toward claim-level verification in LegalQA.

2 Related Work

Research at the intersection of Natural Language Processing and Legal Informatics has rapidly expanded from simple document classification to complex legal reasoning and Retrieval-Augmented Generation (RAG). In this section, we review global LegalQA benchmarks, Vietnamese legal NLP resources, deep legal retrieval (DRILL) methodologies, and recent studies on legal hallucinations.

2.1 Global Legal QA Benchmarks

The evaluation of legal reasoning in language models has evolved from simple multiple-choice bar exam questions to multi-hop legal RAG benchmarks. Early datasets like JEC-QA (Zhong et al., 2020) in Chinese and Bar Exam QA datasets evaluated models on choice selection for statutory knowledge. Modern benchmarks prioritize complex, open-ended reasoning over real-world legal corpora. LegalBench (Guha et al., 2023) introduced a collaboratively constructed benchmark featuring 162 tasks across 6 cognitive legal categories (e.g., issue spotters, rule application, contract interpretation). Recognizing the limitations of static context windows, LegalBench-RAG (Pipitone and Alami, 2024) extended this framework to evaluate retrieval performance over full legal document repositories, emphasizing minimal-span statutory retrieval.

In parallel, domain-specific and multilingual LegalQA datasets have proliferated. LLeQA (Louis et al., 2024) established a French LegalQA dataset focused on legal advice queries over statutory codes. EQUALS (Chen et al., 2023) introduced a real-world Chinese legal QA corpus derived from legal consultation platforms. For Access to Justice (A2J), Li et al. (Li et al., 2024) presented LegalQA for consumer rights and tenant protection, highlighting the need for plain-language responses backed by strict legal authority. More recently, LEXam (Fan et al., 2025) benchmarked legal reasoning across 340 law exams, demonstrating that even frontier LLMs struggle with complex multi-step legal synthesis.

2.2 Vietnamese Legal NLP and Information Retrieval

Compared to English and Chinese, Vietnamese legal NLP historically lacked large-scale benchmark datasets. Early efforts primarily focused on legal text classification or basic keyword matching over legal codes. However, recent benchmarks have significantly bridged this gap. VLQA (Nguyen et al., 2025) introduced a foundational Vietnamese LegalQA dataset comprising 3,129 real-world user queries paired with verified legal grounding across 59,636 statutory documents and articles.

In the realm of legal information retrieval, the DRILL challenge at VLSP 2025 (VLSP Consortium, 2025) established a benchmark for Vietnamese Legal Article Retrieval, pairing over 3,000 legal queries with a corpus of nearly 60,000 legal documents. For visual and document layout understanding, LawViVQA (Nguyen et al., 2024) introduced a visual LegalQA dataset comprising 5,200 document images and 23,000 QA pairs derived from scanned official gazettes. While these benchmarks provide essential building blocks, prior work has evaluated retrieval and generation in isolation, lacking unified end-to-end RAG evaluations with reranking and citation verification.

2.3 Deep Retrieval in the Legal Landscape (DRiLL)

Information retrieval in the legal domain differs substantially from general web search. Legal text is characterized by long document lengths, formal technical vocabulary, cross-referential citation networks, and rigid statutory hierarchies (Abdallah et al., 2023). Deep Retrieval in the expansive Legal Landscape (DRiLL) techniques employ bi-encoder dense retrievers (DPR), ColBERT, and cross-encoder rerankers to capture semantic intent beyond BM25 keyword matching.

Key global datasets supporting DRiLL research include COLIEE for Japanese and Canadian legal article retrieval, LexGLUE (Chalkidis et al., 2022) for multi-task legal understanding, and CUAD (Hendrycks et al., 2021) for expert-annotated legal contract review. For caselaw retrieval, CLERC (Hou et al., 2025) and LePaRD (Mahari et al., 2023) evaluate passage retrieval over millions of federal court opinions, revealing that conventional dense retrievers struggle to achieve high recall on low-frequency statutory citations. For multi-lingual pretraining, MultiLegalPile (Niklaus et al., 2024) provided a 689GB multilingual legal corpus across 24 languages, facilitating legal domain adaptation for base LLMs.

2.4 Legal Hallucinations and AI Reliability

A major barrier to deploying LegalQA systems in professional practice is the propensity of generative models to produce "hallucinated" legal information. Dahl et al. (Dahl et al., 2024) conducted an empirical profiling of legal hallucinations across general-purpose LLMs, discovering hallucination rates exceeding 60% on specific legal reasoning tasks, including the invention of non-existent judicial decisions and fake statutory section numbers.

Furthermore, Magesh et al. (Magesh et al., 2024) evaluated commercial, RAG-based legal AI research platforms (such as Lexis+ AI and Westlaw Precision), reporting that even dedicated legal tools suffer from hallucination rates between 17% and 33%. They observed that models often exhibit "citation misuse"—attaching real, existing legal citations to claims that the cited text does not actually support. This underscores that citation presence is an unreliable proxy for truthfulness, necessitating rigorous human evaluation and claim-level verification protocols.

2.5 Research Gaps Addressed in This Work

While existing literature has advanced Legal RAG and benchmark creation, several critical gaps remain:

1. **Lack of Integrated RAG Benchmarking for Vietnamese:** Prior studies focus either on retrieval alone (DRILL) or static QA (VLQA). There is no end-to-end empirical study measuring the impact of cross-encoder reranking on downstream Vietnamese generation.
2. **Inadequate Citation Audit Mechanisms:** Most RAG frameworks evaluate citations via automated presence or string matching proxies. As established by recent legal hallucination audits (Magesh et al., 2024), string-level proxies over-score citation quality.
3. **Unchecked Memory Leakage:** QA-memory caches are frequently employed to accelerate RAG retrieval, but formal n-gram similarity audits are rarely reported to disprove data contamination.

Our work directly addresses these gaps by evaluating a full Vietnamese LegalQA pipeline with cross-encoder reranking, data leakage auditing, and a multi-axis human evaluation that directly contrasts automatic proxies with human legal judgments.

3 Proposed Methodological Framework

In this section, we present the architecture of our Vietnamese Legal Question Answering system. The framework is designed to retrieve statutory evidence accurately, rerank candidate legal passages using a model-based cross-encoder, and generate legally grounded, auditable answers while mitigating legal hallucinations.

3.1 Corpus Chunking and Legal Metadata Schema

The foundational layer of our Legal RAG pipeline is a structured legal text corpus. Legal documents exhibit rigid structural hierarchies (e.g., Codes, Laws, Decrees, Articles, Clauses, and Points). To preserve semantic integrity without fragmenting legal conditions or exceptions, documents are parsed and segmented into structural chunks.

Each chunk c_i is annotated with a canonical legal citation identifier $\text{CitationID}(c_i)$ and metadata attributes:

$$c_i = \left(\text{Text}_i, \text{CitationID}_i, \text{DocType}_i, \text{DocNum}_i, \text{ArticleNum}_i, \text{ClauseNum}_i \right) \quad (1)$$

Our index comprises ****643,469 corpus chunks**** representing ****261,440 unique citation IDs****, covering major branches of Vietnamese law (civil, criminal, administrative, commercial, and labor law).

3.2 Multi-Stage Retrieval and Cross-Encoder Reranking

Retrieving legal evidence requires balancing broad recall with high precision. Standard dense or sparse retrievers often fail to rank the exact controlling legal clause at top position due to subtle phrasing variations. To resolve this, we implement a two-stage retrieval pipeline:

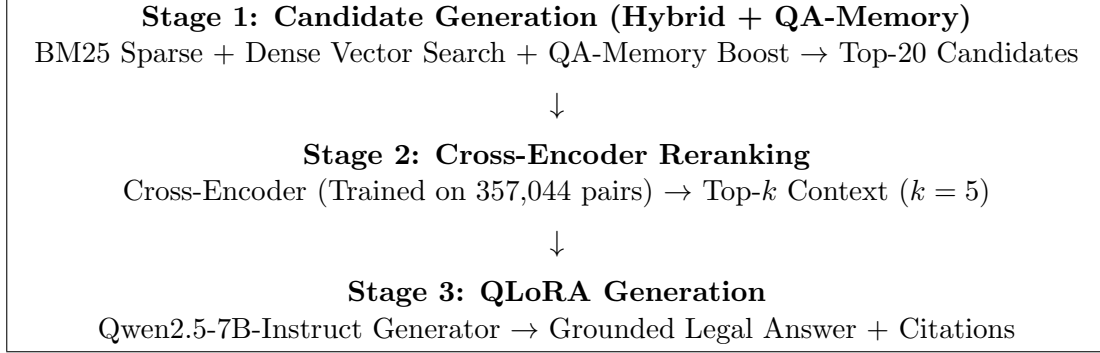


Figure 1: Overview of the Vietnamese LegalQA Retrieval-Reranking-Generation Pipeline.

3.2.1 Stage 1: Hybrid Candidate Generation

Given a user query q , Stage 1 generates a candidate set $\mathcal{C}_{20}$ of top-20 chunks by interpolating BM25 lexical scores $S_{\text{BM25}}(q, c)$, dense bi-encoder embedding similarities $S_{\text{dense}}(q, c)$, and QA-memory relevance scores $S_{\text{mem}}(q, c)$:

$$S_{\text{stage1}}(q, c) = \alpha S_{\text{BM25}}(q, c) + \beta S_{\text{dense}}(q, c) + \gamma S_{\text{mem}}(q, c) \quad (2)$$

where α, β, γ are tuning hyperparameters, and $S_{\text{mem}}(q, c)$ boosts candidate chunks whose statutory citation IDs match historical high-confidence query-citation pairs stored in our QA-memory bank (89,261 records).

3.2.2 Stage 2: Cross-Encoder Reranking

In Stage 2, a cross-encoder model computes full cross-attention between the query q and each candidate chunk $c \in \mathcal{C}_{20}$, producing a refined relevance score $S_{\text{rerank}}(q, c) = \sigma(\text{CrossEncoder}(q, c))$.

To train the cross-encoder, we constructed a dataset of ****357,044 training pairs****. For each positive query-passage pair (q, c^+) , we sampled three hard negative passages (c_1^-, c_2^-, c_3^-) using BM25 and dense retrieval top-rank errors. The cross-encoder is optimized using binary cross-entropy loss:

$$\mathcal{L}_{\text{rerank}} = - \sum_{(q, c, y)} \left[y \log S_{\text{rerank}}(q, c) + (1 - y) \log (1 - S_{\text{rerank}}(q, c)) \right] \quad (3)$$

The top- k reranked chunks ($k = 5$) are selected as the context window $\mathcal{K}_q$ passed to the generator.

3.3 Answer Generation and QLoRA Fine-Tuning

For answer generation, we fine-tune Qwen2.5-7B-Instruct using Quantized Low-Rank Adaptation (QLoRA). Fine-tuning aims to adapt the base LLM to produce structured legal answers that directly address the query, explicitly state the governing statutory citations, and refuse speculation when retrieved evidence is insufficient.

Our Supervised Fine-Tuning (SFT) dataset consists of ****29,145 examples****, split into 26,230 training, 1,457 validation, and 1,458 test examples. QLoRA adapters are attached to the linear

projection matrices of all attention modules, updating less than 1% of total model parameters. The model is trained using causal language modeling loss over target tokens:

$$\mathcal{L}_{\text{SFT}} = - \sum_{t=1}^T \log P_{\theta}(x_t \mid x_{<t}, \mathcal{K}_q, q) \quad (4)$$

3.4 QA-Memory Audit Protocol

Because our Stage 1 retrieval incorporates a QA-memory cache of 89,261 historical query records, there is a potential risk of data contamination between the memory index and the evaluation test set. To ensure scientific validity, we establish a strict n -gram similarity audit protocol.

For each evaluation question q_{eval} , we compute character n -gram Jaccard and cosine similarity against all $q_{\text{mem}} \in \mathcal{M}$:

$$\text{Sim}(q_{\text{eval}}, q_{\text{mem}}) = \frac{\Phi_n(q_{\text{eval}}) \cdot \Phi_n(q_{\text{mem}})}{\|\Phi_n(q_{\text{eval}})\| \|\Phi_n(q_{\text{mem}})\|} \quad (5)$$

Pairs exhibiting similarity ≥ 0.9 are flagged as near-duplicates and analyzed to verify that retrieval gains reflect genuine model generalization rather than memory leakage.

4 Experimental Evaluation and Results

In this section, we present the dataset statistics, retrieval ablation studies, generation performance evaluations, data leakage audits, human verification results, and detailed error analyses of our Vietnamese LegalQA framework.

4.1 Dataset and Artifact Statistics

Our empirical study is conducted on a multi-component legal dataset. Table 1 summarizes the scale of our indexing corpus, training splits, and reranker dataset artifacts.

Table 1: Summary of Dataset and Artifact Statistics for Vietnamese LegalQA.

Artifact / Dataset Split	Count / Volume
Corpus Chunks	643,469
Unique Citation IDs in Index	261,440
QA-Memory Records	89,261
SFT Training Examples	26,230
SFT Validation Examples	1,457
SFT Test Examples	1,458
Reranker Training Records (Positive/Negative Pairs)	357,044

4.2 Retrieval Evaluation and Reranking Ablation

Retrieval performance is evaluated on ****300 held-out legal test queries****. We compare six retrieval configurations using standard Information Retrieval metrics: Recall@ K ($K \in \{1, 5, 10, 20\}$) and Mean Reciprocal Rank (MRR). Results are detailed in Table 2.

The empirical results highlight several key findings:

Table 2: Retrieval Evaluation and Ablation Results across 300 Test Queries.

Retrieval Variant	Recall@1	Recall@5	Recall@10	Recall@20	MRR
BM25 Only	0.2467	0.4267	0.5467	0.6200	0.3370
Dense Vector	0.2900	0.5267	0.6133	0.6867	0.3912
Hybrid (No QA-Memory)	0.2933	0.4933	0.5800	0.6633	0.3821
Hybrid (With QA-Memory)	0.3100	0.5267	0.6200	0.7000	0.4046
Full (Without Model Reranker)	0.3100	0.5267	0.6200	0.7000	0.4046
Full (With Cross-Encoder Reranker)	0.3600	0.6467	0.7033	0.7367	0.4828

- **Dominance of Model-Based Reranking:** Introducing the Stage 2 cross-encoder reranker produces the most substantial single improvement across all metrics. Recall@5 increases from 52.67% to ****64.67%**** (+12.00 percentage points), and MRR jumps from 0.4046 to ****0.4828**** (+19.3% relative gain). This proves that cross-attention reranking effectively promotes the exact statutory evidence chunk to the top ranks.
- **Value of QA-Memory:** Adding QA-memory boosting to hybrid retrieval improves Recall@20 from 66.33% to 70.00% and MRR from 0.3821 to 0.4046, confirming that historical query-citation signals provide effective retrieval prior.

4.3 Answer Generation Experiments

We evaluate answer generation under two experimental settings: (1) *Oracle/SFT Context*, where models are provided with ground-truth evidence passages, and (2) *End-to-End RAG Context*, where models operate on retrieved top-5 context from our Stage 2 cross-encoder reranker. Evaluation is conducted on 200 held-out generation instances. Results are summarized in Table 3.

Table 3: Generation Performance Comparison in Oracle Context and End-to-End RAG Context ($N = 200$).

Variant	Evidence Context	Token F1	ROUGE-L	Citation Pres.	Format Comp.	Faithful. Prox.	Directness Prox.
Extractive Baseline	Oracle/SFT	0.5478	0.4135	0.9300	0.9860	–	0.9150
Qwen Prompt-only	Oracle/SFT	0.4994	0.3896	0.9900	0.0020	–	0.9950
Qwen QLoRA SFT	Oracle/SFT	0.5909	0.4808	0.9900	0.0010	–	0.7400
Qwen QLoRA SFT	Retrieved Top-5	0.5474	0.4092	1.0000	0.0020	0.9408	0.8025

Key observations from generation experiments:

- **Effectiveness of Instruction Tuning:** In oracle context, QLoRA fine-tuning significantly improves answer overlap compared to prompt-only LLM generation, raising Token F1 from 0.4994 to ****0.5909**** and ROUGE-L from 0.3896 to ****0.4808****.
- **RAG Performance Penalty:** Shifting from oracle context to retrieved cross-encoder context reduces Token F1 from 0.5909 to 0.5474 and ROUGE-L from 0.4808 to 0.4092. This gap demonstrates that answer generation quality remains bound by retrieval accuracy.
- **Format Compliance Challenge:** Both prompt-only and QLoRA models exhibited low format compliance (0.0010–0.0020), generating natural language responses rather than strictly adhering to JSON-style schemas.

4.4 QA-Memory Data Leakage Audit

To ensure that QA-memory retrieval gains do not reflect test data contamination, we conducted a data leakage audit comparing 1,458 evaluation test queries against 89,261 QA-memory records.

Table 4 presents the audit metrics.

Table 4: QA-Memory Data Leakage Audit Results.

Audit Metric	Value
Evaluation Test Questions	1,458
QA-Memory Question Records	89,261
Exact Duplicates	0
Normalized Duplicates	0
Near-Duplicates (Character n -gram Cosine ≥ 0.9)	2
Maximum Observed Similarity	0.9388

The audit confirms ****zero exact or normalized duplicates**** between evaluation queries and memory records. Only 2 near-duplicate queries (≥ 0.9 similarity) were identified out of 1,458 queries, confirming that QA-memory performance enhancements represent genuine retrieval augmentation rather than test set memorization.

4.5 Human Evaluation and Metric Discrepancy Analysis

To validate automated proxy metrics, a double-blind preliminary human evaluation was conducted on 75 test instances, with each instance independently scored by two annotators (150 total annotations) across five 5-point Likert scale dimensions. Results are shown in Table 5.

Table 5: Human Evaluation Results across 5 Quality Dimensions (75 Queries, 150 Annotations).

Dimension	Mean	Std	Exact Agree	Within-1 Agree	Pass (≥ 4)	Weak/Fail (≤ 3)
Legal Correctness	4.51	0.85	0.9100	1.0000	0.8800	0.1200
Grounding	3.92	0.78	1.0000	1.0000	0.7600	0.2400
Citation Correctness	3.64	0.48	0.5200	1.0000	0.6400	0.3600
Directness	4.49	0.85	0.9500	1.0000	0.8933	0.1067
Refusal Quality	4.95	0.22	1.0000	1.0000	1.0000	0.0000

4.5.1 The Citation Proxy Discrepancy

The human evaluation uncovers a critical discrepancy between automated proxy metrics and human legal judgment:

- **Automated Citation Proxy Over-Optimism:** Automated metrics report a Citation Presence of ****1.0000**** and a Faithfulness Proxy of ****0.9408****.
- **Human Citation Correctness Deficit:** Human annotators rate Citation Correctness at a mean of only ****3.64/5****, with a high ****Weak/Fail rate of 36.00%**** (≤ 3).

Qualitative inspection reveals that while the model consistently appends legal citation tags to its output, it frequently cites general statutory articles rather than the specific clause or point governing the legal condition. This empirical finding demonstrates that automated citation string matching is insufficient for LegalQA, highlighting the necessity of claim-level legal citation verification.

4.6 Error Analysis

Categorizing failure modes across retrieval and generation pipelines yields actionable insights:

4.6.1 Retrieval Failure Categories

Across 300 retrieval queries, errors fall into two principal categories:

1. **Retrieval Misses (29.67%, 89/300)**: The gold statutory chunk was completely omitted from the top-20 retrieved candidates due to vocabulary mismatch or missing cross-references.
2. **Low Rank After Top-5 (9.00%, 27/300)**: The gold chunk was retrieved in candidate set $\mathcal{C}_{20}$ but ranked below position 5 by the cross-encoder, preventing its inclusion in the generator context.

4.6.2 Generation Failure Categories

For generation, the primary failure mode is ****Format Non-Compliance (100% of generation outputs)****, wherein the model generates natural language explanatory text rather than strictly adhering to schema constraints. Additionally, qualitative case analysis shows that when responding to questions with specific statutory exceptions (e.g., citizenship or residency requirements for civil registry), the model occasionally produces incomplete lists of eligible entities while receiving high automated faithfulness scores, further underscoring proxy metric limitations.

5 Conclusion and Future Directions

In this paper, we presented an architectural framework and comprehensive empirical investigation of a Vietnamese Legal Question Answering system based on Retrieval-Augmented Generation (RAG) and fine-tuned Large Language Models. By indexing 643,469 statutory corpus chunks across 261,440 citation IDs and evaluating on held-out legal queries, we demonstrated that model-based cross-encoder reranking provides substantial retrieval gains, elevating Recall@5 to 64.67% and MRR to 0.4828. Furthermore, QLoRA instruction tuning of Qwen2.5-7B-Instruct achieved strong answer fidelity (Token F1 0.5909, ROUGE-L 0.4808) in oracle context.

Crucially, our work established two fundamental methodological insights for LegalQA. First, a character n -gram data leakage audit verified zero exact or normalized test contamination within our 89,261 QA-memory records. Second, a multi-axis human evaluation exposed a major discrepancy between automated metrics and human legal judgment: automated proxies reported 1.00 citation presence, whereas human legal annotators rated citation correctness at only 3.64/5 with a 36% failure rate, proving that lexical metrics severely mask legal citation misuse.

5.1 Future Research Roadmap

Building upon our empirical findings, we identify five essential research directions to advance trustworthy LegalQA:

1. **Minimal-Span Evidence Retrieval**: Shifting from fixed or article-level chunking to dynamic, minimal statutory span extraction (isolating specific Clauses and Points) to eliminate irrelevant context noise and improve generator precision.
2. **Authority-Aware and Temporal Filtering**: Incorporating explicit statutory hierarchy metadata (Codes > Laws > Decrees > Circulars) and temporal enforcement status (effective dates, amendments, repeals) directly into retriever scoring functions.
3. **Claim-Level Citation Verification**: Developing automated claim-to-evidence verifiers and natural language inference (NLI) models trained specifically on legal entailment to validate that cited statutory sections genuinely support each generated claim prior to output delivery.

4. **Selective QA and Abstention Calibration:** Enhancing confidence calibration so that LegalQA systems reliably refuse to answer or request clarifying facts when retrieved evidence is incomplete, ambiguous, or insufficient.
5. **Neuro-Symbolic Legal Reasoning:** Integrating neural LLM generators with symbolic rule engines and legal knowledge graphs to handle complex multi-hop statutory reasoning involving layered conditions and statutory exceptions.

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